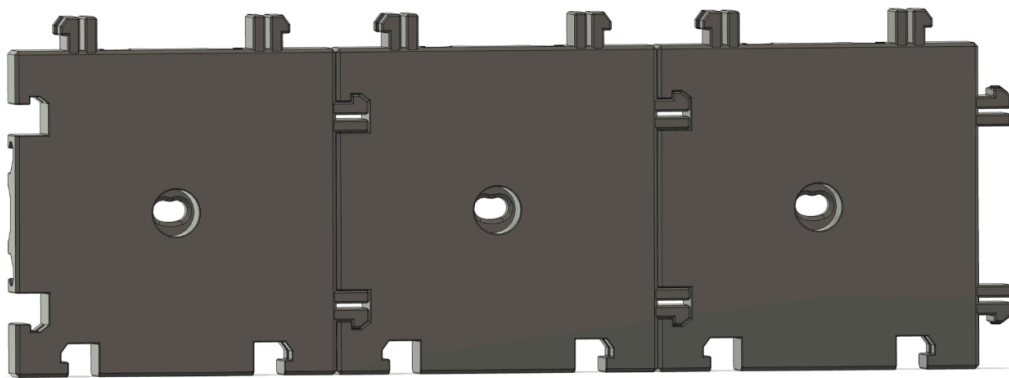




CARGO CONNECT: SPACERS



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file)** – All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file)** – A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files)** – All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print)** – This file is to view the fully assembled model only and not for printing.

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (*affiliate link**) – [**Bambu Lab P1S**](#)

Filaments:

Spacer – Bambu Lab PLA Basic Dark Grey

You can get the filaments we used here (*affiliate link**) – [**Bambu Lab Filament**](#)

*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

Important Information

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The license can be found below the 'Download' icon on each of our model's pages on Thangs.

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Print Settings

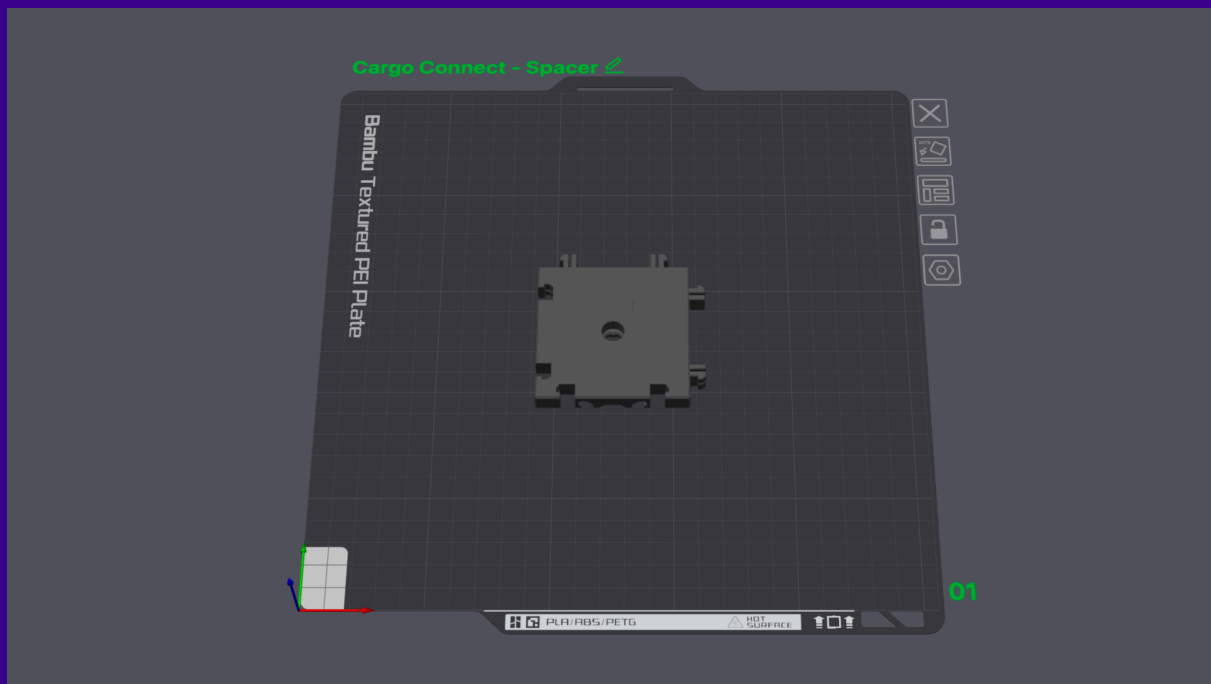
Important Note:

The information detailed in this 'Print Settings' document is the settings we found gave the best results for us. Your printer, materials and printing conditions are unique to you, and due to these widely varying aspects you will know what settings work best for your printers.

Ready to Print Project

A .3mf file named '**1. Ready to Print**' is included in the download. This file is a pre-prepared Bambu Studio project, which includes the settings we used.

(As mentioned above, these settings worked best for us and for our needs, however, you may need to adjust these settings to suit your printing setup and needs)



Individual Part Settings

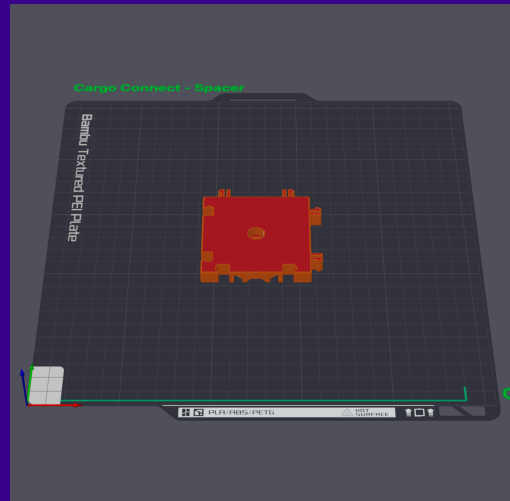
Spacer Tile

Basic Settings:

- **Printer** – Bambu Lab P1S
- **Outer Walls** – 2
- **Infill** – 15%
- **Supports** – None
- **Brim** – Auto

Estimations:

- **Print Time** – 42m 28s
- **Filament Used** – 20.83g



Important Note

As the Spacers do not have the Rails along the top to help with strength, we have added internal structures into the tile to help increase the durability. However, you should test the strength of the Spacers before use to ensure they are durable enough for your specific use case. If needed, increasing the infill and wall loops will boost the strength even more.

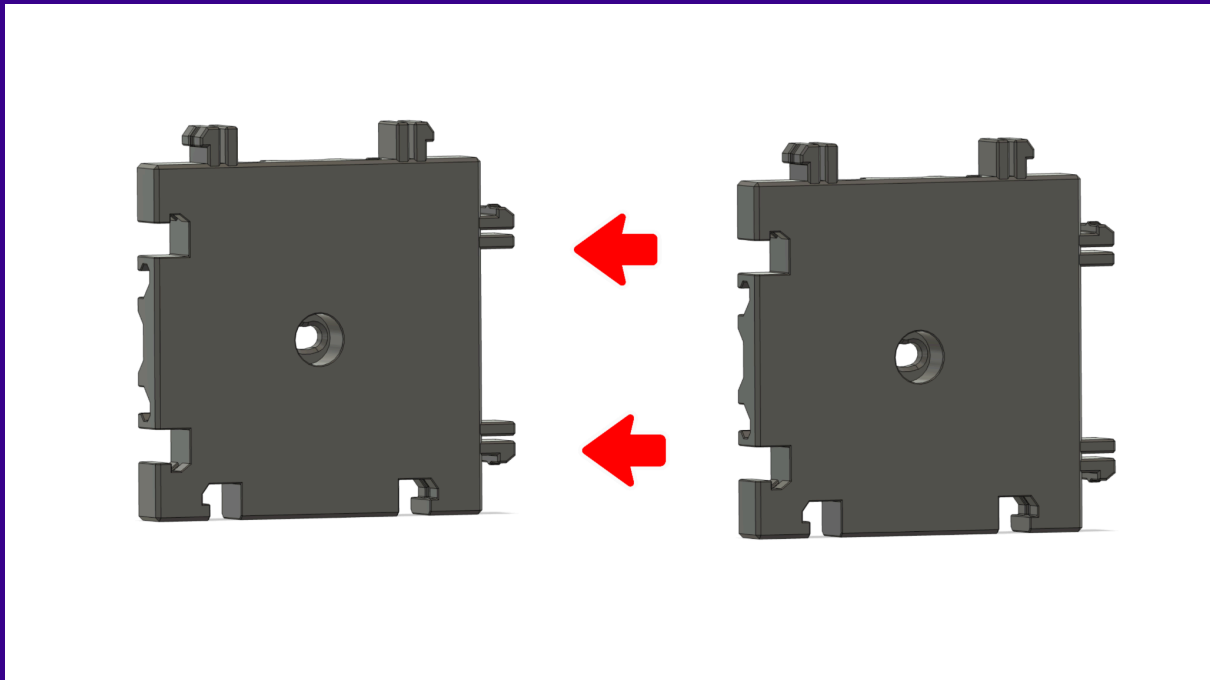
Assembly Guide

Important Note:

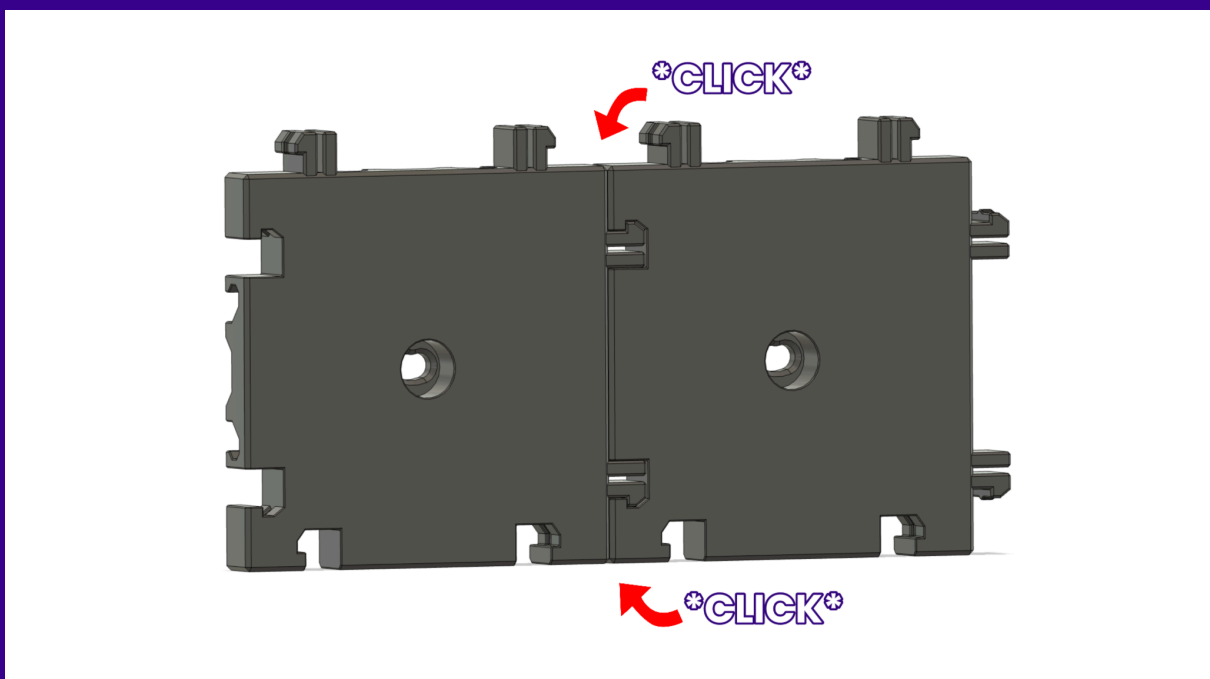
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas – *these may need post processing work or reprinting.*

Step 1

Each tile connects to another tile on any of the four sides using a 'Jigsaw' like connection.



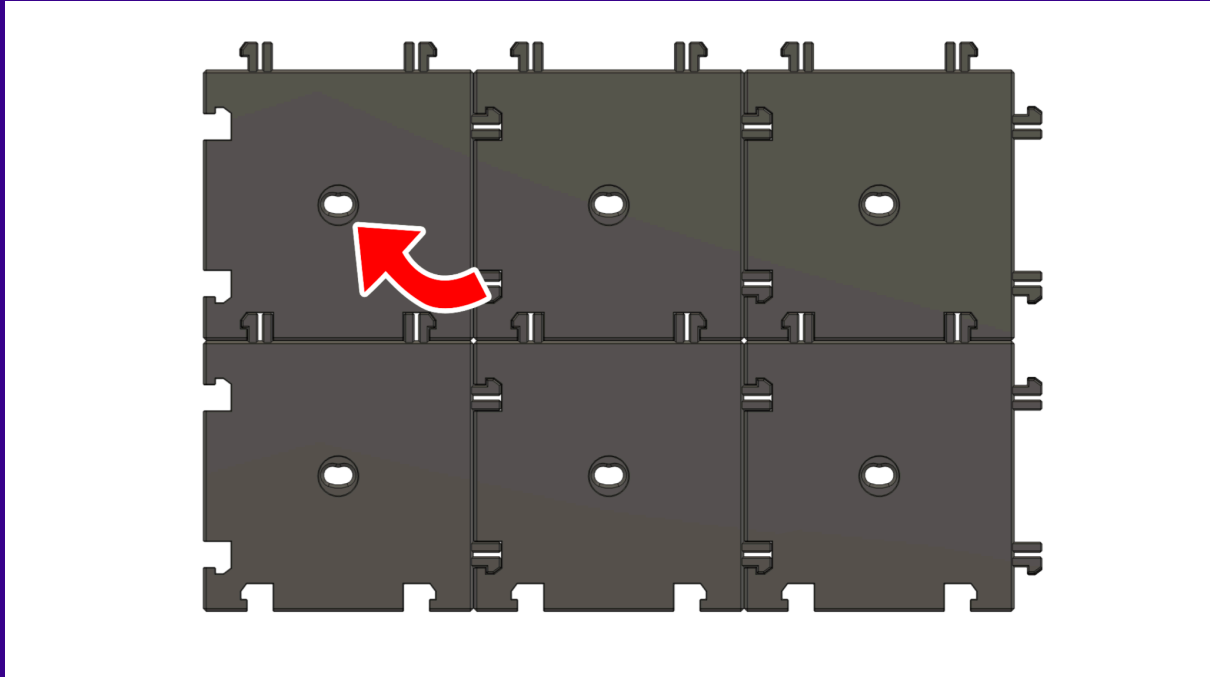
Lay the tiles on a hard flat surface and push them together until you hear the parts click together.



Step 2

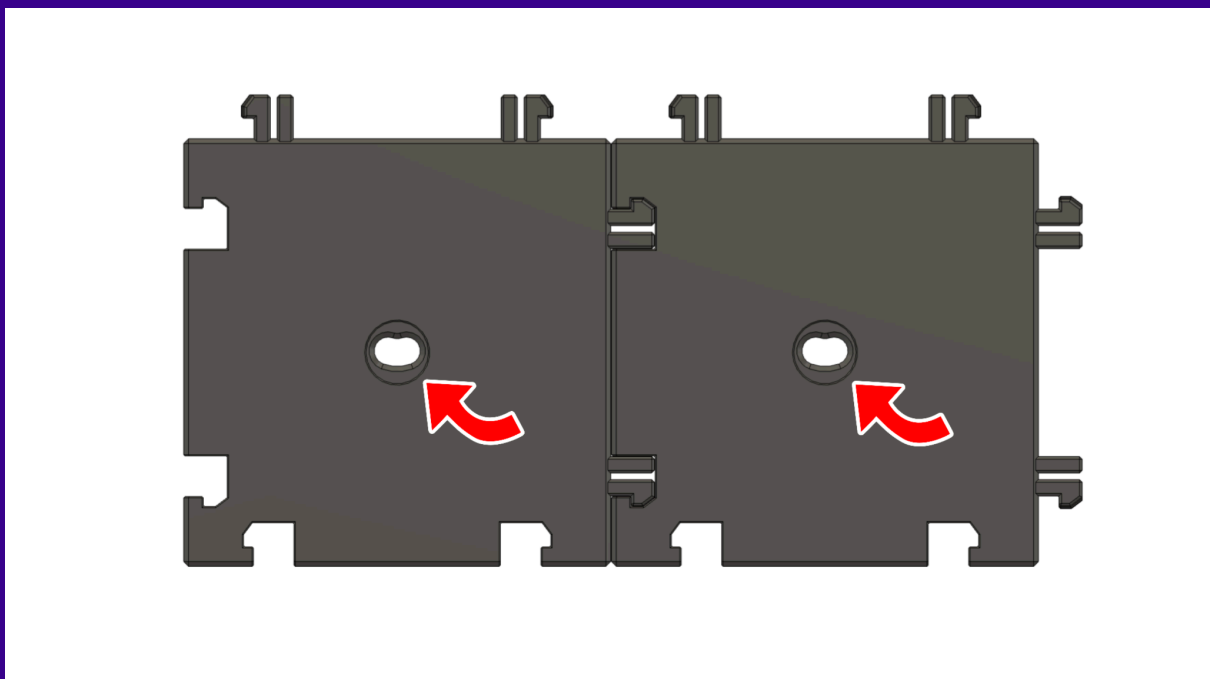
Each tile includes a screw hole for wall mounting. While the interlocking design provides inherent stability, the number and placement of screws used is **entirely at your discretion**, subject to the strength of your wall and the intended weight of your stored items.

You should thoroughly test the stability and strength of your wall and the system before use.



Important Note

As each tile only has one screw hole, you will need to attach at least two tiles together and use the screw hole on each to ensure the tiles do not rotate when in use.

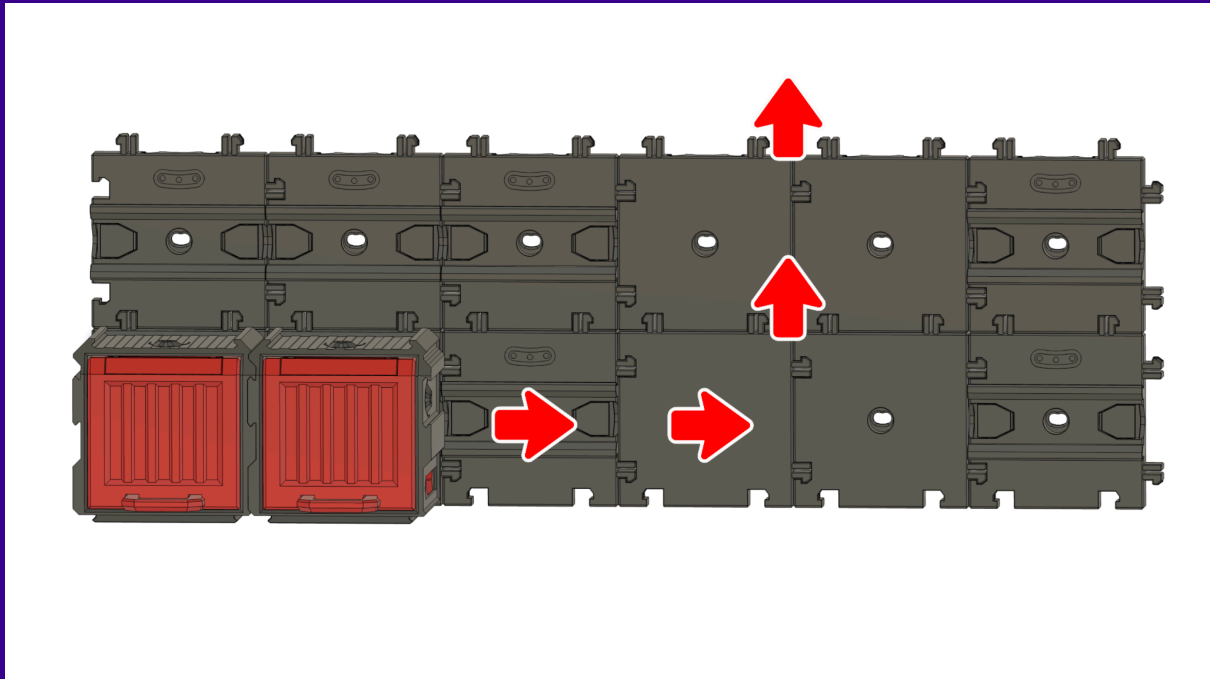


Compatible Components

The Cargo Connect: Spacers are designed to bring ultimate flexibility to your system and are compatible with all other Cargo Connect tiles. Simply use the interlocking pegs around the sides of each tile to connect your desired tiles together.

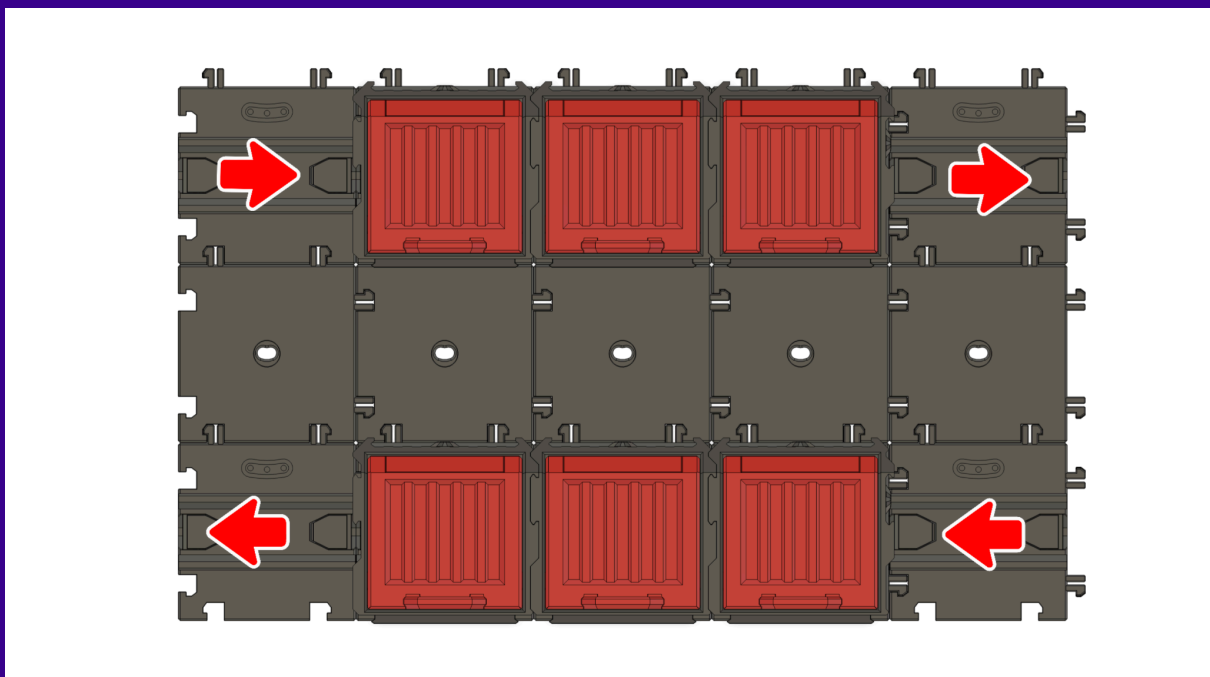
Removing/Adding Cargo Containers

You can add Spacer tiles at certain intervals in your wall system to allow the Cargo Containers to slide off without having to slide them all the way to the end of your system. This is particularly useful for larger systems or systems that are up against another wall.



Adding 'Blank Rows' in between Cargo Containers

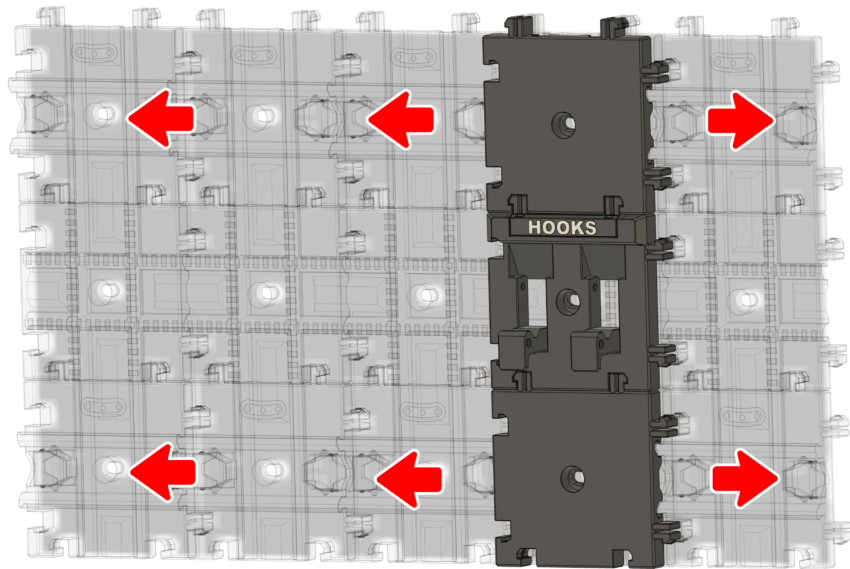
If you want two separate rows of Cargo Containers that don't interlock together, therefore moving separately, the Spacers are a great way to add a blank row so Containers can move freely past each other.



Make functional tiles compatible with the 'Cargo Connect: Rails'

Certain functional tiles, like the Cargo Connect: Hooks, are not able to slide along the Cargo Connect: Rails due to their built in features.

Using the specifically designed tracks cut into the back of the Cargo Connect: Spacers, they can be added to tiles like the Cargo Connect: Hooks to make them fully compatible with the sliding rails.



Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur – below are some potential solutions that may be useful:

Warping and Bed Adhesion

Poor bed adhesion or inconsistent room temperatures can cause the print to warp off the build plate during printing.

The Cargo Connect tiles should come perfectly flat off your print bed, if they do not this may affect functionality like the interlocking pegs and sliding along the rails. Using a brim or an appropriate glue for your build plate may help with this.

Tiles not sliding smoothly along Rails

Depending on your print quality and filament used, you may feel some friction when sliding the components along the rails. Firstly, check for any print imperfections or lumps of plastic that need removing. If the friction persists, we found adding a small amount of lubricant to the rails then sliding an attachment back and forth a few times helps this drastically (*Be sure to use lubricant suitable for your chosen filament*).

If Problems Continue:

If the above has not resolved any issues, we recommend printing a variety of comprehensive tolerance tests on your printer. You can download and print these online from various reputable websites such as Thangs.

These tests will help determine the capabilities of your specific printer and where improvements may be needed..